

ALEXANDER STOKES

SOFTWARE ENGINEERING STUDENT

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PERSONAL SUMMARY

- Interest in real-world applications of software, networking, and automation
- Passion for improving efficiency and problem-solving
- Experienced IT Technician & Hardware / Software Technician (current)
- Won the Waikato x Cardiff University Cybersecurity CTF competition
- 3rd place (tertiary category) and 10th overall for the NZ Cybersecurity Competition 2022

TECHNICAL SKILLS

Javascript

- Developed NodeJS packages, APIs, a chrome extension chat application, and more.
- Skilled in Sockets, MQTT brokers, ES6 modules, and basic Express usage.

Java

- Developed RegEx compiler, LZW compression algorithm, OpenCV retina comparison tool and various Android apps.
- Skilled in Java SOAP, OpenCV, and Java Swing.

C#

- Developed various automation tools, and control systems for clients.
- Skilled in using hardware drivers (TecomC4, ICT, Crestron) and Topshelf.

Git

- Used git in all commercial, university and personal projects.
- Skilled in the usage of Git & version controls – Branch managing, cherry-picking, rebasing, etc.

Other languages and technologies used:

- Python, C, C++, PHP, PostgreSQL, Strapi, and Bash.

WORK EXPERIENCE

IT Hardware & Software Technician

March, 2022 – Present

Arc Automation, Security & Electrical, Hamilton

- Development of custom control software for custom hardware systems
- Programming & maintenance of security systems (alarms, cameras, gates, etc.)
- Automating business and residential tasks

Key skills:

- Time Management – Skillfully working around deadlines, while meeting client expectations
- Learning – Constantly expanding on knowledge base
- Code cleanliness & documentation – Ensuring code is readable and easily maintainable

Software Engineer Intern

November, 2022 – March, 2023

SquareKicker

- Backend development and database management
- Code base migration
- Mentored a junior developer

Key skills:

- Agile scrum – Team coordination and task management
- Strapi – Primary technology used for the product

IT Technician

April, 2019 – March, 2022

Millennium Technology & Support, Hamilton

- Physical device repairs
- On-site customer assistance
- Remote support for businesses

Key skills:

- Organisation – Managing products, customer devices and job timelines
- Professionalism – Always maintaining a professional manner when interacting with clients

QUALIFICATIONS

Bachelor of Software Engineering (Honours)

University of Waikato, Hamilton

2021–Present

PROJECT EXPERIENCE

Forklift Danger System for Arc Automation, Security & Electrical

- NodeJS control system
- Uses GPS locations, generated from AI object detection cameras
- Detects danger between forklifts and people
- Includes a custom settings interface, email notifications, and a map to visualise points

Key tools/skills:

- NodeJS – Web Sockets, MQTT, Express, and more
- C++ – RTSP Stream, and Fiducial Recognition

Automatic Tag Scanner and Writer for Arc Automation, Security & Electrical

- Raspberry Pico controlled components with electronics
- Computer Vision and OCR text reading from key fobs

Key tools/skills:

- Python – OCR, hardware interaction, control system
- OpenCV – Image processing

Other Projects

- RegEx compiler
- Competency Management System
- A* Pathfinding Algorithm
- Engineer contract deadline calculator
- Umpire allocations assistive tool
- Sierpinski Triangle – Demo visible via Sierpinski.Skwangles.com

Key tools/skills:

- Organisation – Skillfully managing deadlines, teams, and project files and fitting to strict specifications
- Code documentation & cleanliness – Ensuring code can be easily understood, and subsequently debugged
- Communication – Effectively communicating with teammates to ensure projects progress

INTERESTS

My life is a captivating blend of roles and passions. As a dedicated community-qualified umpire, I officiate A-grade secondary school and reserve club hockey, ensuring fair play. Additionally, I volunteer as a lighting technician at Activate Church, where I create engaging visual backgrounds. Language learning is another strong interest of mine, and through engaging with people from diverse cultures, I've developed a self-taught conversational Danish fluency over the past three years.

ACHEIVEMENTS & AWARDS

- Winner – Waikato x Cardiff University CTF Competition
- 3rd Place (Tertiary Category) – NZ Cybersecurity Competition 2022 (10th Overall)
- Brian Perry Charitable Trust Scholarship recipient
- Waikato University Computer Science Undergraduate Scholarship recipient

REFERENCES

Dimitri Mucznik, Supervisor, Arc Automation, Security, & Electrical, 027 451 2254

Nick Ippolito, Founder & CEO, Squarekicker, 027 826 0021

Barry Mitchell, Former Managing Director, Millennium Technology & Support, 021 589 191